

Well-posedness theory for coupled multidimensional quadratic BSDEs in portfolio equilibria

W4411801764_p0 (website page)

Website #18 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

We are asked for *general* well-posedness results for multidimensional quadratic BSDE systems

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s,$$

where Y is $\mathbb{R}^m$ -valued, Z is $\mathbb{R}^{m \times d}$ -valued, the generator has quadratic growth in Z , and the components may be *genuinely coupled*: f^i may depend on all rows of Z , including cross terms $Z^j \cdot Z^k$ with $j \neq k$. The target solution class is the natural bounded/BMO class:

$$Y \in \mathbb{S}^\infty, \quad Z \in \mathbb{H}_{\text{BMO}}^2.$$

The strongest literal reading is:

For arbitrary bounded terminal condition ξ and arbitrary genuinely coupled quadratic generator f satisfying only a quadratic-growth condition, prove existence and uniqueness of a bounded/BMO solution.

Its negation is:

There exist m, d, T , a bounded terminal condition ξ , and a genuinely coupled quadratic generator f for which no bounded/BMO solution exists, or uniqueness fails.

I will first attack this strongest reading and disprove it by an explicit counterexample. I then prove a positive theorem: under an explicit *small-data* condition, one gets existence and uniqueness of a bounded/BMO solution *without any diagonal, triangular, or Markovian structure*.

Alternative readings considered

The prompt can be read in two ways.

- **Hardest reading:** obtain a large-data unrestricted theorem for all fully coupled quadratic generators.
- **Weaker reading:** give the strongest rigorous theorem/obstruction available.

I adopt the hardest reading first. Theorem 1 below shows that this literal unrestricted program is false. I then retreat, explicitly and honestly, to a sharp partial theorem: a fully coupled small-data existence/uniqueness result in $\mathbb{S}^\infty \times \mathbb{H}_{\text{BMO}}^2$.

Related literature found

- **Pardoux–Peng (1990)** — BACKGROUND. Foundational L^2 -well-posedness theorem for Lipschitz BSDEs; useful as the baseline comparison point from which quadratic theory departs. (sciencedirect.com)

- **Tevzadze (2008)** — PARTIAL. Establishes existence/uniqueness for multidimensional quadratic BSDEs under a *small terminal condition* assumption via a BMO/contraction argument; this is the classical small-data paradigm. (sciencedirect.com)
- **Frei–dos Reis (2011)** — PARTIAL. Gives an explicit counterexample showing that, unlike the scalar case, bounded terminal data do *not* guarantee a bounded solution for multidimensional quadratic BSDEs. (research.ed.ac.uk)
- **Cheridito–Nam (2015)** — PARTIAL. Proves solvability for special classes: Markovian/projectable structures and strictly subquadratic growth. (research.monash.edu)
- **Hu–Tang (2016)** — PARTIAL. Develops local/global solvability for *diagonally quadratic* systems, where the i -th quadratic term depends only on the i -th row of Z . (econpapers.repec.org)
- **Xing–Žitković (2018)** — PARTIAL. Gives a wide but still structural *Markovian* solvability theory for quadratic BSDE systems. (eprints.lse.ac.uk)
- **Harter–Richou (2019)** — PARTIAL. Proves existence/uniqueness for a class characterized by stability and a priori estimates of approximants; again, additional structure is essential. (oskar-bordeaux.fr)
- **Jackson–Žitković (2022)** — PARTIAL. Solves a non-Markovian *triangular* class and explicitly frames the general fully coupled large-data theory as inaccessible by current techniques; it also identifies a matrix-BMO obstruction to a naive Girsanov approach. (par.nsf.gov)
- **Fan–Hu–Tang (2023)** — PARTIAL. Broadens the diagonal theory, but still within diagonal/strictly diagonal regimes or related one-sided structural hypotheses. (sciencedirect.com)

No consulted source claimed a global large-data theorem for arbitrary fully coupled non-Markovian cross-term generators in the bounded/BMO class; the sources I found still rely on smallness, diagonal/triangular structure, Markovianity, or other special assumptions. (par.nsf.gov)

Proof sketch

There are really two separate mathematical messages.

First, the unrestricted problem is *false*. I exhibit an extremely simple two-dimensional, one-Brownian-motion, time-homogeneous polynomial generator

$$f(z^1, z^2) = \left(\frac{(z^1)^2 - (z^2)^2}{2}, z^1 z^2 \right),$$

which is fully quadratic and genuinely coupled. Under the identification $\mathbb{R}^2 \cong \mathbb{C}$, this is just the complex square $v \mapsto \frac{1}{2}v^2$. That algebraic form is special enough that the exponential transform $X_t = e^{Y_t^1 + iY_t^2}$ linearizes the BSDE. Choosing a bounded terminal condition $\xi = (0, \pi \mathbf{1}_{\{W_T \geq 0\}})$, one gets $X_T \in \{-1, 1\}$ with mean 0, hence $X_0 = 0$, impossible because $X_0 = e^{Y_0^1 + iY_0^2} \neq 0$. This disproves the strongest reading.

Second, although unrestricted large-data well-posedness is impossible, one can still prove a general *fully coupled* small-data theorem. The right hypotheses are not diagonality or triangularity, but an abstract quadratic-Lipschitz bound

$$|f(t, y, z) - f(t, y', z')| \leq \beta |y - y'| + \gamma(|z| + |z'|)|z - z'|.$$

This allows completely arbitrary cross-coupling in Z .

The proof is a Picard contraction in the norm

$$N(Y, Z) := \max \left\{ \|Y\|_\infty, \frac{1}{\sqrt{2}} \|Z\|_{\text{BMO}} \right\}.$$

For a frozen pair (U, V) , define $g_t = f(t, U_t, V_t)$ and solve the linear BSDE

$$Y_t = \xi + \int_t^T g_s ds - \int_t^T Z_s dW_s.$$

The key estimates are:

1. a conditional L^1 -estimate

$$\sup_\tau \left\| \mathbb{E}_\tau \int_\tau^T |g_s| ds \right\|_\infty \leq \alpha T + \beta T N(U, V) + 2\gamma N(U, V)^2;$$

2. an elementary BMO moment lemma showing that $V \in \text{BMO}$ implies $\int_0^T |V_s|^2 ds \in L^2$, so the frozen equation is well-defined;
3. a linear BSDE estimate that turns the previous conditional L^1 -bound into an N -bound.

Choosing the ball radius as the smaller positive root of

$$2\gamma R^2 - (1 - \beta T)R + \delta = 0, \quad \delta := \|\xi\|_\infty + \alpha T,$$

makes the Picard map both invariant and contractive. This yields existence and uniqueness *in that ball*, with no componentwise structural restriction on the quadratic coupling.

Main result

Write

$$\|Y\|_\infty := \text{ess sup}_{(\omega, t) \in \Omega \times [0, T]} |Y_t(\omega)|,$$

and, for predictable $Z: \Omega \times [0, T] \rightarrow \mathbb{R}^{m \times d}$,

$$\|Z\|_{\text{BMO}} := \sup_{\tau \leq T} \left\| \mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \right\|_\infty^{1/2},$$

where the supremum runs over all stopping times $\tau \leq T$, and $|\cdot|$ is the Euclidean/Frobenius norm.

Theorem 1 (No unrestricted bounded/BMO well-posedness). *Fix any $T > 0$. Let $m = 2$, $d = 1$, and define*

$$f(z^1, z^2) = \left(\frac{(z^1)^2 - (z^2)^2}{2}, z^1 z^2 \right), \quad (z^1, z^2) \in \mathbb{R}^2.$$

Then $|f(z)| = \frac{1}{2}|z|^2$, so f has genuine quadratic growth and includes a cross term.

Let

$$\xi := (0, \pi \mathbf{1}_{\{W_T \geq 0\}}).$$

Then the BSDE

$$Y_t = \xi + \int_t^T f(Z_s) ds - \int_t^T Z_s dW_s$$

admits no solution with $Y \in \mathbb{S}^\infty(\mathbb{R}^2)$. In particular it admits no solution in the natural class

$$Y \in \mathbb{S}^\infty(\mathbb{R}^2), \quad Z \in \mathbb{H}_{\text{BMO}}^2(\mathbb{R}^{2 \times 1}).$$

Theorem 2 (Small-data well-posedness for arbitrary quadratic coupling). *Assume, in addition, that the filtration $\mathbb{F}$ has the martingale representation property with respect to W (for example, $\mathbb{F}$ is the usual augmentation of the Brownian filtration).*

Let $f: \Omega \times [0, T] \times \mathbb{R}^m \times \mathbb{R}^{m \times d} \rightarrow \mathbb{R}^m$ be progressively measurable, and suppose there exist constants $\alpha, \beta, \gamma \geq 0$ such that for all $(t, \omega, y, y', z, z')$,

$$|f(t, \omega, 0, 0)| \leq \alpha,$$

and

$$|f(t, \omega, y, z) - f(t, \omega, y', z')| \leq \beta|y - y'| + \gamma(|z| + |z'|)|z - z'|.$$

No diagonal, triangular, or Markovian structure is assumed.

Set

$$\delta := \|\xi\|_\infty + \alpha T.$$

Assume

$$\beta T < 1, \quad D := (1 - \beta T)^2 - 8\gamma\delta > 0.$$

Define

$$R_* := \frac{1 - \beta T - \sqrt{D}}{4\gamma}$$

(with the convention $R_* = 0$ if $\delta = 0$).

Then the BSDE

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s$$

has a solution (Y, Z) such that

$$\max \left\{ \|Y\|_\infty, \frac{1}{\sqrt{2}} \|Z\|_{\text{BMO}} \right\} \leq R_*.$$

Moreover, this solution is unique among all solutions satisfying the same bound.

Proof

Throughout the proof, all norms on $\mathbb{R}^m$ and $\mathbb{R}^{m \times d}$ are Euclidean/Frobenius norms.

Proof of Theorem 1. Assume, for contradiction, that there exists a bounded adapted solution (Y, Z) , with

$$Y = (Y^1, Y^2), \quad Z = (Z^1, Z^2), \quad Z^1, Z^2 \in \mathbb{R}.$$

Identify $\mathbb{R}^2$ with $\mathbb{C}$ by setting

$$U_t := Y_t^1 + iY_t^2, \quad V_t := Z_t^1 + iZ_t^2.$$

Because

$$\frac{1}{2}V_t^2 = \frac{1}{2}((Z_t^1)^2 - (Z_t^2)^2) + i Z_t^1 Z_t^2,$$

the BSDE is exactly

$$U_t = i\pi \mathbf{1}_{\{W_T \geq 0\}} + \int_t^T \frac{1}{2} V_s^2 ds - \int_t^T V_s dW_s.$$

Equivalently, in differential form,

$$dU_t = -\frac{1}{2} V_t^2 dt + V_t dW_t.$$

Now define

$$X_t := e^{U_t}.$$

Applying Itô's formula to the smooth map $u \mapsto e^u$ on $\mathbb{C} \cong \mathbb{R}^2$,

$$dX_t = X_t dU_t + \frac{1}{2} X_t d\langle U \rangle_t.$$

Since $d\langle U \rangle_t = V_t^2 dt$, we get

$$dX_t = X_t \left(-\frac{1}{2} V_t^2 dt + V_t dW_t \right) + \frac{1}{2} X_t V_t^2 dt = X_t V_t dW_t.$$

Hence X is a complex-valued local martingale.

Because Y is bounded, Y^1 is bounded, and therefore

$$|X_t| = |e^{Y_t^1 + iY_t^2}| = e^{Y_t^1}$$

is bounded. Thus both the real and imaginary parts of X are bounded real local martingales, hence true martingales. Therefore X itself is a true martingale.

At terminal time,

$$X_T = e^{i\pi \mathbf{1}_{\{W_T \geq 0\}}} = \mathbf{1}_{\{W_T < 0\}} - \mathbf{1}_{\{W_T \geq 0\}}.$$

Since W_T is centered Gaussian,

$$\mathbb{P}(W_T \geq 0) = \mathbb{P}(W_T < 0) = \frac{1}{2},$$

so

$$\mathbb{E}[X_T] = 0.$$

Because X is a martingale,

$$X_0 = \mathbb{E}[X_T] = 0.$$

But

$$X_0 = e^{U_0} \neq 0,$$

because the complex exponential never vanishes. This contradiction proves the theorem. $\square$

For the positive theorem, define

$$N(Y, Z) := \max \left\{ \|Y\|_\infty, \frac{1}{\sqrt{2}} \|Z\|_{\text{BMO}} \right\}.$$

Lemma 1 (BMO moments of $\int |Z|^2$). *Let $V \in \mathbb{H}_{\text{BMO}}^2(\mathbb{R}^{m \times d})$, and for $t \in [0, T]$ set*

$$A_t := \int_t^T |V_s|^2 ds.$$

Then for every integer $n \geq 1$ and every stopping time $\tau \leq T$,

$$\mathbb{E}_\tau[A_\tau^n] \leq n! \|V\|_{\text{BMO}}^{2n}.$$

In particular,

$$\int_0^T |V_s|^2 ds \in L^n(\mathbb{P}) \quad \text{for every } n \geq 1.$$

Proof. We argue by induction on n .

For $n = 1$, the claim is exactly the definition of $\|V\|_{\text{BMO}}$:

$$\mathbb{E}_\tau[A_\tau] = \mathbb{E}_\tau \left[\int_\tau^T |V_s|^2 ds \right] \leq \|V\|_{\text{BMO}}^2.$$

Assume the estimate true for $n-1$, with $n \geq 2$. Since A_t is absolutely continuous with derivative $\frac{d}{dt}A_t = -|V_t|^2$, we have

$$A_\tau^n = n \int_\tau^T A_s^{n-1} |V_s|^2 ds.$$

Taking conditional expectation at time τ ,

$$\mathbb{E}_\tau[A_\tau^n] = n \mathbb{E}_\tau \left[\int_\tau^T A_s^{n-1} |V_s|^2 ds \right].$$

Because $|V_s|^2$ is $\mathcal{F}_s$ -measurable, the tower property yields

$$\mathbb{E}_\tau[A_\tau^n] = n \mathbb{E}_\tau \left[\int_\tau^T \mathbb{E}_s[A_s^{n-1}] |V_s|^2 ds \right].$$

By the induction hypothesis applied at time s ,

$$\mathbb{E}_s[A_s^{n-1}] \leq (n-1)! \|V\|_{\text{BMO}}^{2n-2}.$$

Therefore

$$\mathbb{E}_\tau[A_\tau^n] \leq n! \|V\|_{\text{BMO}}^{2n-2} \mathbb{E}_\tau \left[\int_\tau^T |V_s|^2 ds \right] \leq n! \|V\|_{\text{BMO}}^{2n}.$$

This closes the induction. □

Lemma 2 (Linear BSDE estimate in $\mathbb{S}^\infty \times \text{BMO}$). *Let (Y, Z) solve*

$$Y_t = \eta + \int_t^T g_s ds - \int_t^T Z_s dW_s,$$

where $\eta \in L^\infty(\Omega; \mathbb{R}^m)$, and

$$\Gamma(g) := \sup_{\tau \leq T} \left\| \mathbb{E}_\tau \left[\int_\tau^T |g_s| ds \right] \right\|_\infty < \infty.$$

Then

$$N(Y, Z) \leq \|\eta\|_\infty + \Gamma(g).$$

In particular, if $\eta = 0$, then

$$N(Y, Z) \leq \Gamma(g).$$

Proof. First,

$$Y_t = \mathbb{E}_t \left[\eta + \int_t^T g_s ds \right].$$

Hence

$$|Y_t| \leq \|\eta\|_\infty + \mathbb{E}_t \left[\int_t^T |g_s| ds \right] \leq \|\eta\|_\infty + \Gamma(g),$$

so

$$\|Y\|_\infty \leq \|\eta\|_\infty + \Gamma(g).$$

Now fix a stopping time $\tau \leq T$. Applying Itô's formula to $|Y_t|^2$ on $[\tau, T]$,

$$|Y_\tau|^2 + \int_\tau^T |Z_s|^2 ds = |\eta|^2 + 2 \int_\tau^T Y_s \cdot g_s ds - 2 \int_\tau^T Y_s \cdot Z_s dW_s.$$

Taking $\mathbb{E}_\tau$,

$$\mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] = \mathbb{E}_\tau[|\eta|^2] - |Y_\tau|^2 + 2 \mathbb{E}_\tau \left[\int_\tau^T Y_s \cdot g_s ds \right].$$

Discarding the nonpositive term $-|Y_\tau|^2$,

$$\mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \leq \|\eta\|_\infty^2 + 2\|Y\|_\infty \mathbb{E}_\tau \left[\int_\tau^T |g_s| ds \right].$$

Taking essential suprema and using the previous bound on $\|Y\|_\infty$,

$$\|Z\|_{\text{BMO}}^2 \leq \|\eta\|_\infty^2 + 2(\|\eta\|_\infty + \Gamma(g))\Gamma(g).$$

Since

$$\|\eta\|_\infty^2 + 2(\|\eta\|_\infty + \Gamma(g))\Gamma(g) \leq 2(\|\eta\|_\infty + \Gamma(g))^2,$$

we obtain

$$\frac{1}{\sqrt{2}}\|Z\|_{\text{BMO}} \leq \|\eta\|_\infty + \Gamma(g).$$

Together with the $\|Y\|_\infty$ -bound, this proves

$$N(Y, Z) \leq \|\eta\|_\infty + \Gamma(g).$$

The special case $\eta = 0$ is immediate. □

Proof of Theorem 2. Let

$$\mathcal{X} := \mathbb{S}^\infty(\mathbb{R}^m) \times \mathbb{H}_{\text{BMO}}^2(\mathbb{R}^{m \times d}),$$

endowed with the norm N . This is a Banach space, so its closed ball

$$\mathcal{B}_R := \{(U, V) \in \mathcal{X} : N(U, V) \leq R\}$$

is complete for every $R \geq 0$.

We first note that the hypotheses imply the quadratic growth bound

$$|f(t, \omega, y, z)| \leq |f(t, \omega, 0, 0)| + |f(t, \omega, y, z) - f(t, \omega, 0, 0)| \leq \alpha + \beta|y| + \gamma|z|^2.$$

We now define the Picard map. Fix $(U, V) \in \mathcal{B}_R$, and set

$$g_t := f(t, U_t, V_t).$$

Then

$$|g_t| \leq \alpha + \beta|U_t| + \gamma|V_t|^2 \leq \alpha + \beta R + \gamma|V_t|^2.$$

Because $N(U, V) \leq R$, we have $\|V\|_{\text{BMO}} \leq \sqrt{2}R$. Hence Lemma 1 with $n = 2$ gives

$$\mathbb{E} \left[\left(\int_0^T |V_s|^2 ds \right)^2 \right] \leq 2! (\sqrt{2}R)^4 = 8R^4.$$

Therefore

$$\mathbb{E} \left[\left(\int_0^T |g_s| ds \right)^2 \right] \leq 3(\alpha T)^2 + 3(\beta T R)^2 + 3\gamma^2 \mathbb{E} \left[\left(\int_0^T |V_s|^2 ds \right)^2 \right] < \infty.$$

So the random vector

$$H := \xi + \int_0^T g_s ds$$

belongs to $L^2(\Omega; \mathbb{R}^m)$.

Since $\mathbb{F}$ has the martingale representation property with respect to W , there exists a unique predictable $Z \in \mathbb{H}^2(\mathbb{R}^{m \times d})$ such that

$$\mathbb{E}_t[H] = \mathbb{E}[H] + \int_0^t Z_s dW_s.$$

Define

$$Y_t := \mathbb{E}_t[H] - \int_0^t g_s ds.$$

Then, because $\mathbb{E}_T[H] = H$,

$$Y_t = H - \int_0^t g_s ds - \int_t^T Z_s dW_s = \xi + \int_t^T g_s ds - \int_t^T Z_s dW_s.$$

Thus (Y, Z) solves the frozen BSDE

$$Y_t = \xi + \int_t^T f(s, U_s, V_s) ds - \int_t^T Z_s dW_s.$$

We define $\Phi(U, V) := (Y, Z)$.

Step 1: Φ maps $\mathcal{B}_R$ into itself for $R = R_$.*

For any stopping time $\tau \leq T$,

$$\mathbb{E}_\tau \left[\int_\tau^T |g_s| ds \right] \leq \alpha T + \beta T \|U\|_\infty + \gamma \mathbb{E}_\tau \left[\int_\tau^T |V_s|^2 ds \right].$$

Hence

$$\Gamma(g) \leq \alpha T + \beta T R + \gamma \|V\|_{\text{BMO}}^2 \leq \alpha T + \beta T R + 2\gamma R^2.$$

By Lemma 2,

$$N(\Phi(U, V)) \leq \|\xi\|_\infty + \Gamma(g) \leq \delta + \beta T R + 2\gamma R^2.$$

Now choose $R = R_*$. By definition of R_* ,

$$2\gamma R_*^2 - (1 - \beta T)R_* + \delta = 0,$$

which is equivalent to

$$\delta + \beta T R_* + 2\gamma R_*^2 = R_*.$$

Therefore

$$N(\Phi(U, V)) \leq R_*,$$

so $\Phi(\mathcal{B}_{R_*}) \subseteq \mathcal{B}_{R_*}$.

Step 2: Φ is a strict contraction on $\mathcal{B}_{R_}$.*

Take $(U, V), (\tilde{U}, \tilde{V}) \in \mathcal{B}_{R_*}$, and let

$$(\Delta Y, \Delta Z) := \Phi(U, V) - \Phi(\tilde{U}, \tilde{V}).$$

Then $(\Delta Y, \Delta Z)$ solves

$$\Delta Y_t = \int_t^T \Delta g_s ds - \int_t^T \Delta Z_s dW_s,$$

where

$$\Delta g_s := f(s, U_s, V_s) - f(s, \tilde{U}_s, \tilde{V}_s).$$

By the quadratic-Lipschitz hypothesis,

$$|\Delta g_s| \leq \beta |U_s - \tilde{U}_s| + \gamma(|V_s| + |\tilde{V}_s|) |V_s - \tilde{V}_s|.$$

Fix a stopping time $\tau \leq T$. Then

$$\mathbb{E}_\tau \left[\int_\tau^T |\Delta g_s| ds \right] \leq \beta T \|U - \tilde{U}\|_\infty + \gamma \mathbb{E}_\tau \left[\int_\tau^T (|V_s| + |\tilde{V}_s|) |\Delta V_s| ds \right].$$

By conditional Cauchy–Schwarz,

$$\mathbb{E}_\tau \left[\int_\tau^T (|V_s| + |\tilde{V}_s|) |\Delta V_s| ds \right] \leq \left(\mathbb{E}_\tau \left[\int_\tau^T (|V_s| + |\tilde{V}_s|)^2 ds \right] \right)^{1/2} \left(\mathbb{E}_\tau \left[\int_\tau^T |\Delta V_s|^2 ds \right] \right)^{1/2}.$$

Taking essential suprema,

$$\Gamma(\Delta g) \leq \beta T \|U - \tilde{U}\|_\infty + \gamma (\|V\|_{\text{BMO}} + \|\tilde{V}\|_{\text{BMO}}) \|\Delta V\|_{\text{BMO}}.$$

Since $(U, V), (\tilde{U}, \tilde{V}) \in \mathcal{B}_{R_*}$,

$$\|V\|_{\text{BMO}} + \|\tilde{V}\|_{\text{BMO}} \leq 2\sqrt{2} R_*, \quad \|\Delta V\|_{\text{BMO}} \leq \sqrt{2} N((U, V) - (\tilde{U}, \tilde{V})).$$

Therefore

$$\Gamma(\Delta g) \leq (\beta T + 4\gamma R_*) N((U, V) - (\tilde{U}, \tilde{V})).$$

Now Lemma 2, applied with terminal condition 0, yields

$$N(\Phi(U, V) - \Phi(\tilde{U}, \tilde{V})) \leq \Gamma(\Delta g) \leq (\beta T + 4\gamma R_*) N((U, V) - (\tilde{U}, \tilde{V})).$$

Because

$$4\gamma R_* = 1 - \beta T - \sqrt{D},$$

we get

$$\beta T + 4\gamma R_* = 1 - \sqrt{D} < 1.$$

So Φ is a strict contraction on the complete metric space $\mathcal{B}_{R_*}$.

By the contraction mapping principle, Φ has a unique fixed point in $\mathcal{B}_{R_*}$. That fixed point is exactly a solution (Y, Z) of the original BSDE satisfying

$$N(Y, Z) \leq R_*.$$

Uniqueness in $\mathcal{B}_{R_*}$ follows from the same contraction estimate. $\square$

Gap to the full statement

The full problem, read literally as a large-data theorem for *all* fully coupled quadratic generators and bounded terminal conditions, is false by Theorem 1. So there is no “gap” in the sense of a missing proof of that literal strongest reading: it cannot be true.

The remaining gap is therefore a *reformulation gap*. What remains open is to identify broad, checkable hypotheses — substantially weaker than diagonal/triangular structure, but stronger than mere quadratic growth — under which one can prove:

- existence for *large* bounded terminal conditions;
- uniqueness in the *entire* bounded/BMO class, not only in a small ball;
- results without the auxiliary martingale-representation assumption on the filtration;
- continuation/pasting principles that survive genuinely coupled cross terms.

Theorem 2 shows that structural assumptions are *not* needed for small-data solvability; Theorem 1 shows that *some* additional restriction is logically indispensable.

Self red-team

1. **Counterexample search against Theorem 2.** I first checked the explicit generator from Theorem 1. It satisfies the quadratic-Lipschitz assumption with $\alpha = \beta = 0$ and $\gamma = \frac{1}{2}$, but for the chosen terminal condition $\delta = \|\xi\|_\infty = \pi$, so

$$D = 1 - 8 \cdot \frac{1}{2} \cdot \pi = 1 - 4\pi < 0.$$

Hence Theorem 2 does *not* incorrectly predict existence.

2. **Counterexample search against Lemma 1.** For deterministic $V_t \equiv c$, one has

$$A_\tau = c^2(T - \tau),$$

and the bound $\mathbb{E}_\tau[A_\tau^n] \leq n! \|V\|_{\text{BMO}}^{2n}$ is valid with room to spare. So the lemma is consistent on the simplest test family.

3. **Counterexample search against Lemma 2.** For $g \equiv 0$, the lemma reduces to the obvious bound $N(Y, Z) \leq \|\eta\|_\infty$. For $\eta = 0$ and deterministic bounded g , the estimate becomes $N(Y, Z) \leq \Gamma(g)$, again sharp in the right scale.
4. **Three plausible referee objections.**

- (a) “The complex notation in Theorem 1 may hide a mistake.” I checked this explicitly: $\frac{1}{2}(Z^1 + iZ^2)^2 = \frac{1}{2}((Z^1)^2 - (Z^2)^2) + iZ^1Z^2$, exactly the given generator.
- (b) “The frozen linear BSDE may not be well-defined because $f(U, V)$ is only quadratic in V .” This is why Lemma 1 is included: it gives

$$\int_0^T |V_s|^2 ds \in L^2,$$

hence $H = \xi + \int_0^T f(U, V) ds \in L^2$, so the martingale representation step is legitimate.

- (c) “*The uniqueness claim is too strong.*” I do *not* claim full-class uniqueness. The theorem states uniqueness only inside the explicit small ball $\mathcal{B}_{R_*}$, and the proof proves exactly that.

5. **Boundary cases checked.**

- If $\delta = 0$, then $R_* = 0$ and the unique solution is the zero solution.
- If $D = 0$, the contraction constant becomes 1; the proof no longer yields uniqueness. The theorem therefore correctly requires $D > 0$.
- Theorem 1 works for every $T > 0$, so shrinking the horizon alone cannot rescue arbitrary large bounded terminal values.

6. **Minimality of the negative example.** I deliberately minimized dimensions and algebraic complexity: $m = 2$, $d = 1$, time-homogeneous generator, polynomial quadratic nonlinearity, and a two-point bounded terminal condition.

Ideas tried

1. **Global contraction on the whole bounded/BMO space.** Idea: run Picard iteration in $\mathbb{S}^\infty \times \mathbb{H}_{\text{BMO}}^2$ without any smallness. Obstruction: the quadratic difference estimate necessarily carries a factor proportional to $\|Z\|_{\text{BMO}} + \|\tilde{Z}\|_{\text{BMO}}$, so no global contraction can hold. Theorem 1 shows this is not just a proof defect but a genuine impossibility. Next step: only a different structural mechanism (monotonicity, Lyapunov function, PDE regularity, etc.) could replace smallness.
2. **Short-time local solvability for arbitrary bounded terminal data.** Idea: perhaps very small time horizon alone makes every fully coupled quadratic system solvable. Obstruction: Theorem 1 already fails on *every* horizon $T > 0$. So shortness of the interval, by itself, is insufficient in the fully coupled setting. Next step: one would need additional regularity/splitting assumptions on the terminal condition, not merely small T .
3. **Weighted norms to remove the restriction $\beta T < 1$.** Idea: replace $\|Y\|_\infty$ by an exponentially weighted sup norm to turn the y -Lipschitz factor βT into β/λ . Obstruction: the bookkeeping becomes much uglier because the data terms $\|\xi\|_\infty$ and αT then acquire exponential factors $e^{\lambda T}$. I did not obtain a clean final statement whose constants were both transparent and unquestionably correct, so I did not promote that approach into the main theorem. Next step: carry out the weighted-norm variant carefully; it should plausibly yield a theorem on arbitrary horizons but with exponentially small data.
4. **Girsanov linearization for full-class uniqueness.** Idea: given two solutions, linearize the difference equation and absorb the Z -terms by a change of measure. Obstruction: in the genuinely coupled vector case the linearized coefficient is matrix-valued, and one needs strong reverse-Hölder type control of a matrix stochastic exponential; under the present assumptions I could not justify that. Next step: add structural assumptions guaranteeing a matrix reverse-Hölder property, or work in a class where the linearized coefficient is scalar/triangular.

Invoked results

No externally sourced theorem is used as an internal proof step. The proof is self-contained modulo:

- **Contraction mapping principle** (T00L, standard). *Statement.* A strict contraction on a complete metric space has a unique fixed point. *Hypothesis audit.* In the proof of Theorem 2, the space $\mathcal{B}_{R_*}$ is a closed subset of the Banach product space $\mathbb{S}^\infty \times \mathbb{H}_{\text{BMO}}^2$, hence complete; the map Φ is shown explicitly to satisfy

$$N(\Phi(x) - \Phi(y)) \leq q N(x - y), \quad q = 1 - \sqrt{D} < 1.$$

- **Martingale representation property** (assumed, not invoked from an external source). *Role.* It is an explicit hypothesis of Theorem 2, used to represent the square-integrable martingale $t \mapsto \mathbb{E}_t[H]$ as a stochastic integral against W .

Self-confidence

3 — I checked every derivation and constant in the two proved theorems and could not break them; the main remaining caution is not about correctness, but about scope: the positive theorem is intentionally a small-ball result rather than full-class uniqueness.

Bibliography

```
@article{PardouxPeng1990,
  author = {Pardoux, Etienne and Peng, Shige},
  title = {Adapted Solution of a Backward Stochastic Differential Equation},
  journal = {Systems \& Control Letters},
  volume = {14},
  number = {1},
  pages = {55--61},
  year = {1990},
  doi = {10.1016/0167-6911(90)90082-6}
}

@article{Tevzadze2008,
  author = {Tevzadze, Revaz},
  title = {Solvability of backward stochastic differential equations with quadratic growth},
  journal = {Stochastic Processes and their Applications},
  volume = {118},
  number = {3},
  pages = {503--515},
  year = {2008},
  doi = {10.1016/j.spa.2007.05.009}
}

@article{FreiDosReis2011,
  author = {Frei, Christoph and dos Reis, Gon{\c c}alo},
  title = {A financial market with interacting investors: does an equilibrium exist?},
  journal = {Mathematics and Financial Economics},
  volume = {4},
  number = {3},
  pages = {161--182},
  year = {2011},
  doi = {10.1007/s11579-011-0039-0}
}

@article{CheriditoNam2015,
  author = {Cheridito, Patrick and Nam, Kihun},
```

```

    title = {Multidimensional quadratic and subquadratic BSDEs with special structure},
    journal = {Stochastics},
    volume = {87},
    number = {5},
    pages = {871--884},
    year = {2015},
    doi = {10.1080/17442508.2015.1013959}
}

@article{HuTang2016,
  author = {Hu, Ying and Tang, Shanjian},
  title = {Multi-dimensional backward stochastic differential equations of diagonally quadratic generators},
  journal = {Stochastic Processes and their Applications},
  volume = {126},
  number = {4},
  pages = {1066--1086},
  year = {2016},
  doi = {10.1016/j.spa.2015.10.011}
}

@article{XingZitkovic2018,
  author = {Xing, Hao and {\v Z}itkovi{\c}, Gordan},
  title = {A class of globally solvable Markovian quadratic BSDE systems and applications},
  journal = {The Annals of Probability},
  volume = {46},
  number = {1},
  pages = {491--550},
  year = {2018},
  doi = {10.1214/17-AOP1190}
}

@article{HarterRichou2019,
  author = {Harter, Jonathan and Richou, Adrien},
  title = {A stability approach for solving multidimensional quadratic BSDEs},
  journal = {Electronic Journal of Probability},
  volume = {24},
  pages = {Paper No. \ 4, 51 pp.},
  year = {2019},
  doi = {10.1214/18-EJP260}
}

@article{JacksonZitkovic2022,
  author = {Jackson, Joe and {\v Z}itkovi{\c}, Gordan},
  title = {Existence and Uniqueness for Non-Markovian Triangular Quadratic BSDEs},
  journal = {SIAM Journal on Control and Optimization},
  volume = {60},
  number = {3},
  pages = {1642--1666},
  year = {2022},
  doi = {10.1137/21M1435689}
}

@article{FanHuTang2023,
  author = {Fan, Shengjun and Hu, Ying and Tang, Shanjian},
  title = {Multi-dimensional backward stochastic differential equations of diagonally quadratic generators: The general result},
  journal = {Journal of Differential Equations},
  volume = {368},

```

```
pages = {105--140},  
year = {2023},  
doi = {10.1016/j.jde.2023.05.041}  
}
```